

```

import pandas as pd
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import warnings
warnings.filterwarnings("ignore")
import matplotlib.pyplot as plt
import numpy as np

```

```

z = pd.read_csv(r"C:\Users\skj_h\OneDrive\Desktop\Uber dataset.csv")
z

```

	trip_duration	distance_traveled	num_of_passengers	fare
tip \				
0	748.0	2.75	1.0	75.00
24				
1	1187.0	3.43	1.0	105.00
24				
2	730.0	3.12	1.0	71.25
0				
3	671.0	5.63	3.0	90.00
0				
4	329.0	2.09	1.0	45.00
12				
...	...	...	...	...
..				
209668	1617.0	8.42	1.0	150.00
47				
209669	438.0	1.29	1.0	48.75
12				
209670	571.0	2.82	1.0	63.75
0				
209671	491.0	2.16	1.0	56.25
0				
209672	3614.0	33.72	1.0	337.50
0				
	miscellaneous_fees	total_fare	surge_applied	
0	6.300	105.300	0	
1	13.200	142.200	0	
2	26.625	97.875	1	
3	9.750	99.750	0	
4	13.200	70.200	0	
...	...	...	...	
209668	5.800	202.800	0	
209669	34.575	95.325	1	
209670	6.000	69.750	0	
209671	13.500	69.750	0	
209672	2.250	339.750	0	

```
[209673 rows x 8 columns]
```

```
z.isnull().sum()
```

```
trip_duration      0
distance_traveled  0
num_of_passengers  0
fare               0
tip               0
miscellaneous_fees 0
total_fare         0
surge_applied      0
dtype: int64
```

```
z.shape
```

```
(209673, 8)
```

```
z.size
```

```
1677384
```

```
z.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 209673 entries, 0 to 209672
Data columns (total 8 columns):
```

#	Column	Non-Null Count	Dtype
0	trip_duration	209673 non-null	float64
1	distance_traveled	209673 non-null	float64
2	num_of_passengers	209673 non-null	float64
3	fare	209673 non-null	float64
4	tip	209673 non-null	int64
5	miscellaneous_fees	209673 non-null	float64
6	total_fare	209673 non-null	float64
7	surge_applied	209673 non-null	int64

```
dtypes: float64(6), int64(2)
```

```
memory usage: 12.8 MB
```

```
z.dtypes
```

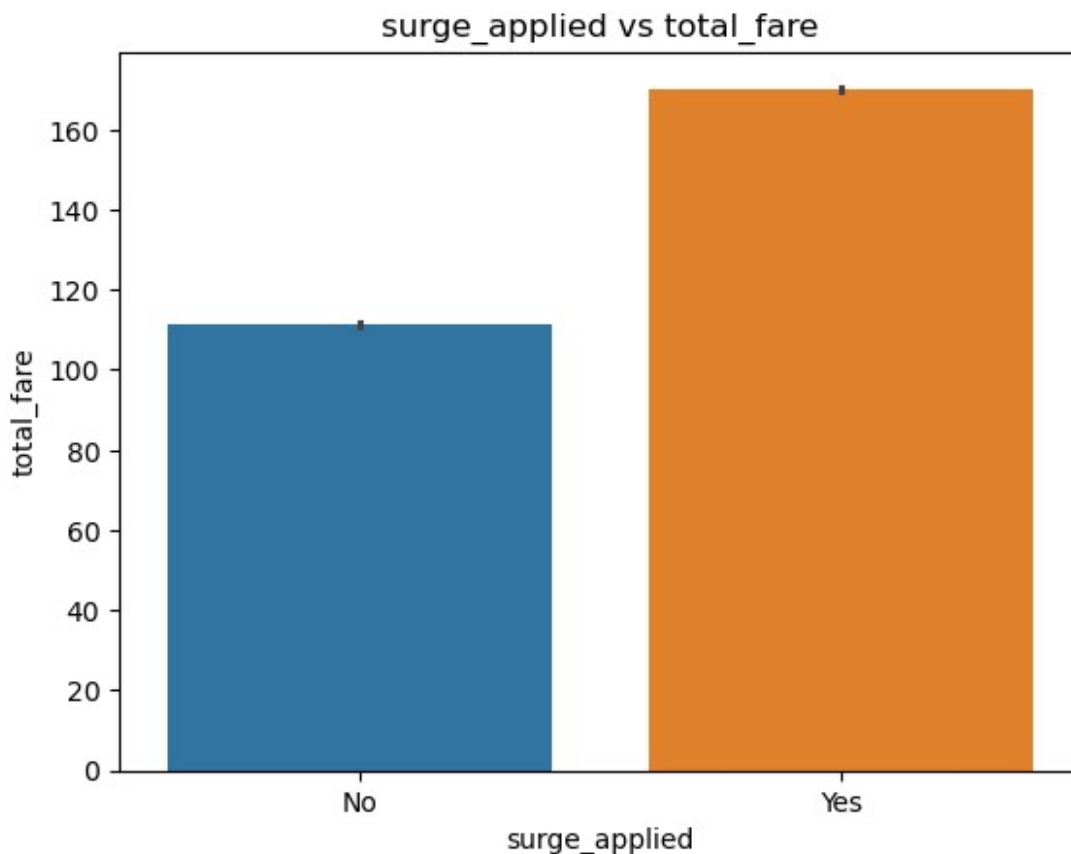
```
trip_duration      float64
distance_traveled  float64
num_of_passengers  float64
fare               float64
tip               int64
miscellaneous_fees float64
total_fare         float64
surge_applied      int64
dtype: object
```

```
def change_string(a):
    if(a == 1):
        return("Yes")
    else:
        return("No")

z["surge_applied"] = z["surge_applied"].apply(change_string)
z["surge_applied"] = z["surge_applied"].astype(str)

plt.title("surge_applied vs total_fare")
sns.barplot(x = z["surge_applied"], y = z["total_fare"], data = z, hue
= z["surge_applied"])

<Axes: title={'center': 'surge_applied vs total_fare'},
xlabel='surge_applied', ylabel='total_fare'>
```



```
b = z.copy()
b.drop(["surge_applied"], axis = 1, inplace = True)
b

   trip_duration  distance_traveled  num_of_passengers  fare
tip \
```

0	748.0	2.75	1.0	75.00
24				
1	1187.0	3.43	1.0	105.00
24				
2	730.0	3.12	1.0	71.25
0				
3	671.0	5.63	3.0	90.00
0				
4	329.0	2.09	1.0	45.00
12				
...	...	...	...	...
..				
209668	1617.0	8.42	1.0	150.00
47				
209669	438.0	1.29	1.0	48.75
12				
209670	571.0	2.82	1.0	63.75
0				
209671	491.0	2.16	1.0	56.25
0				
209672	3614.0	33.72	1.0	337.50
0				

	miscellaneous_fees	total_fare
0	6.300	105.300
1	13.200	142.200
2	26.625	97.875
3	9.750	99.750
4	13.200	70.200
...	...	...
209668	5.800	202.800
209669	34.575	95.325
209670	6.000	69.750
209671	13.500	69.750
209672	2.250	339.750

[209673 rows x 7 columns]

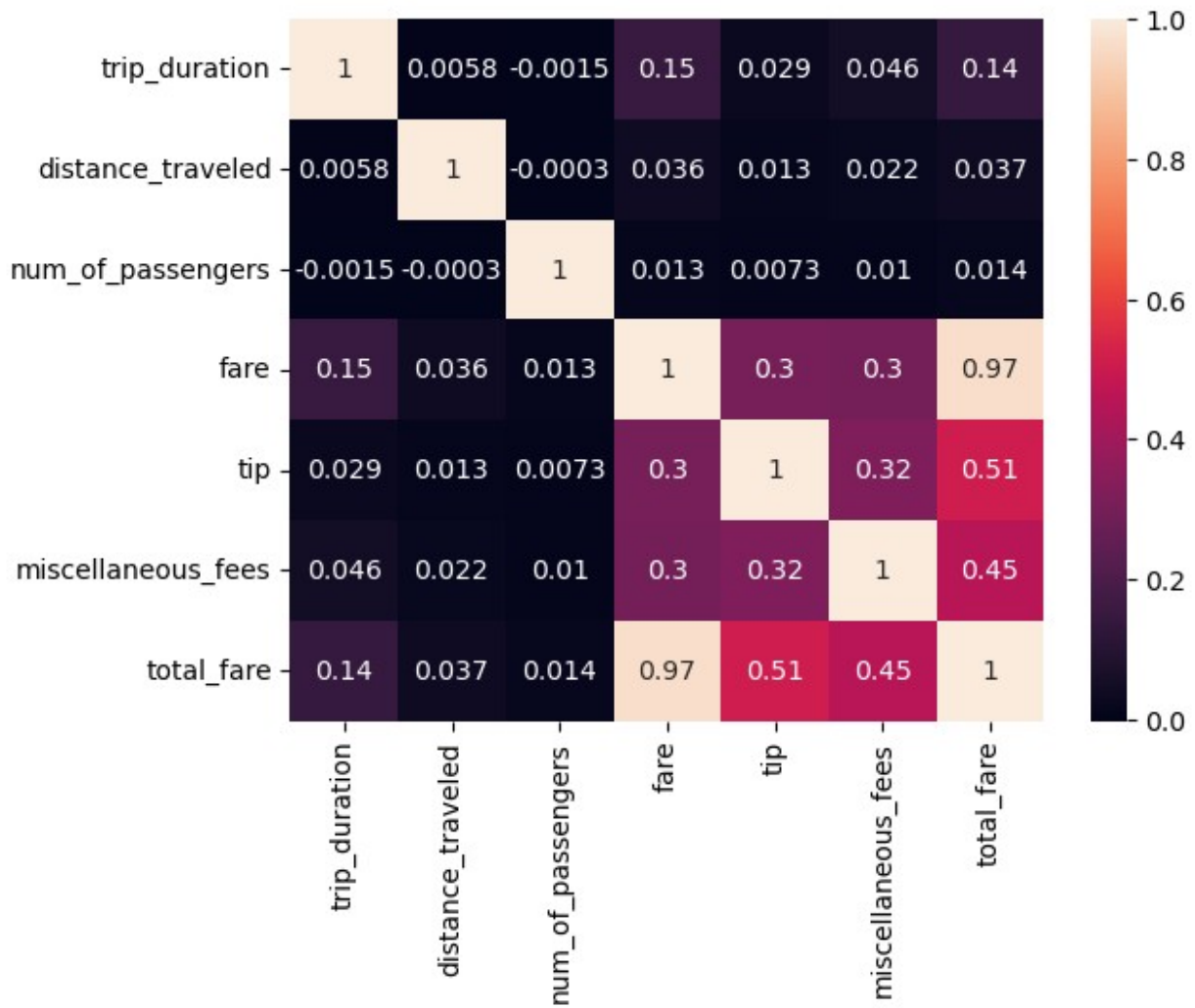
```
b.corr()["total_fare"].sort_values(ascending = False)
```

total_fare	1.000000
fare	0.966748
tip	0.508639
miscellaneous_fees	0.452568
trip_duration	0.142159
distance_traveled	0.036677
num_of_passengers	0.014234

Name: total_fare, dtype: float64

```
sns.heatmap(b.corr(), annot = True)
```

<Axes: >



```
z.columns
```

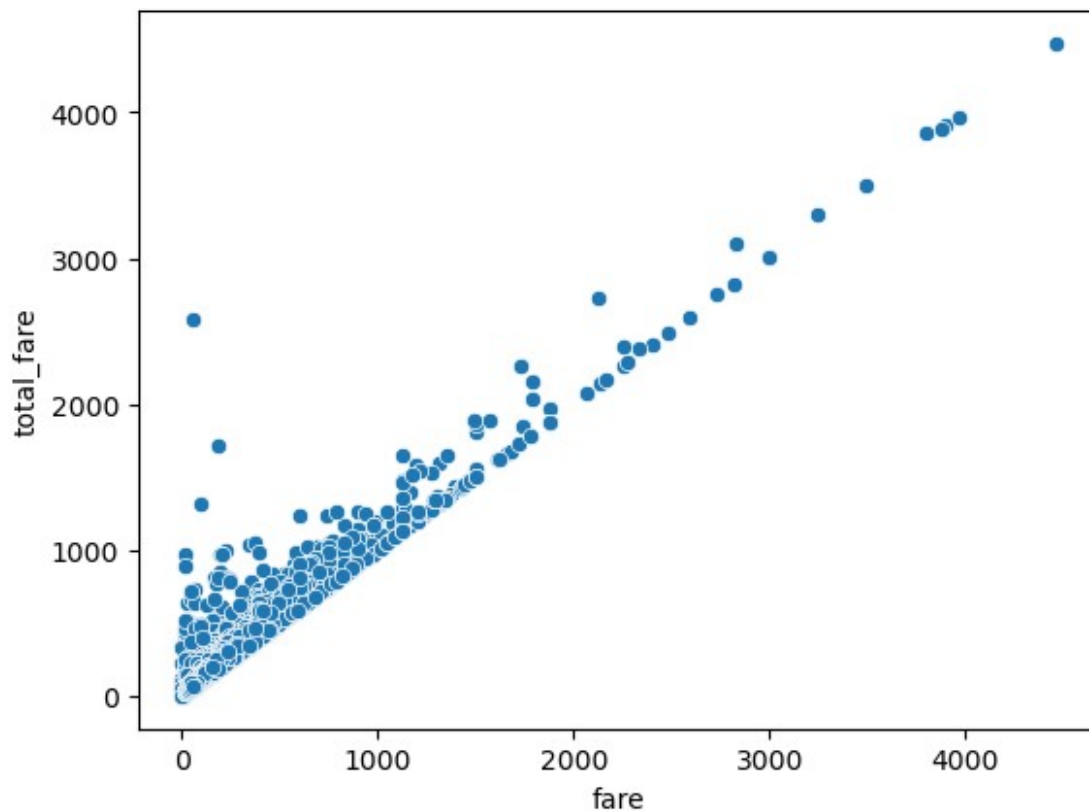
```
Index(['trip_duration', 'distance_traveled', 'num_of_passengers',  
      'fare',  
      'tip', 'miscellaneous_fees', 'total_fare', 'surge_applied'],  
      dtype='object')
```

```
z5 = z.groupby(["surge_applied"])["num_of_passengers"].count()
```

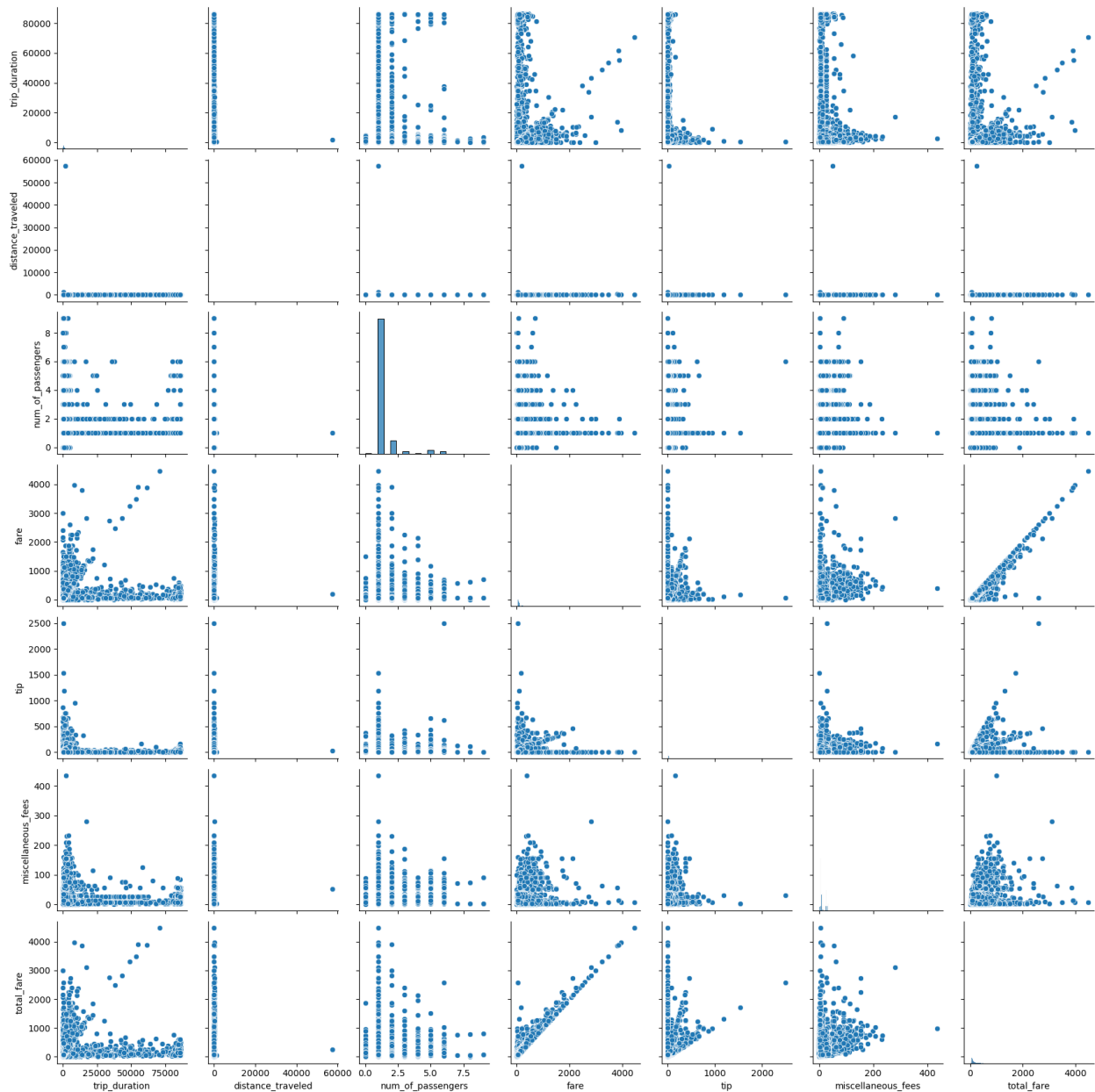
```
pd.DataFrame(z5)
```

```
      num_of_passengers  
surge_applied  
No                150855  
Yes                58818
```

```
z.columns  
Index(['trip_duration', 'distance_traveled', 'num_of_passengers',  
      'fare',  
      'tip', 'miscellaneous_fees', 'total_fare', 'surge_applied'],  
      dtype='object')  
  
sns.scatterplot(x = z["fare"], y = z["total_fare"], data = z)  
<Axes: xlabel='fare', ylabel='total_fare'>
```



```
sns.pairplot(z)  
<seaborn.axisgrid.PairGrid at 0x2d141946d50>
```

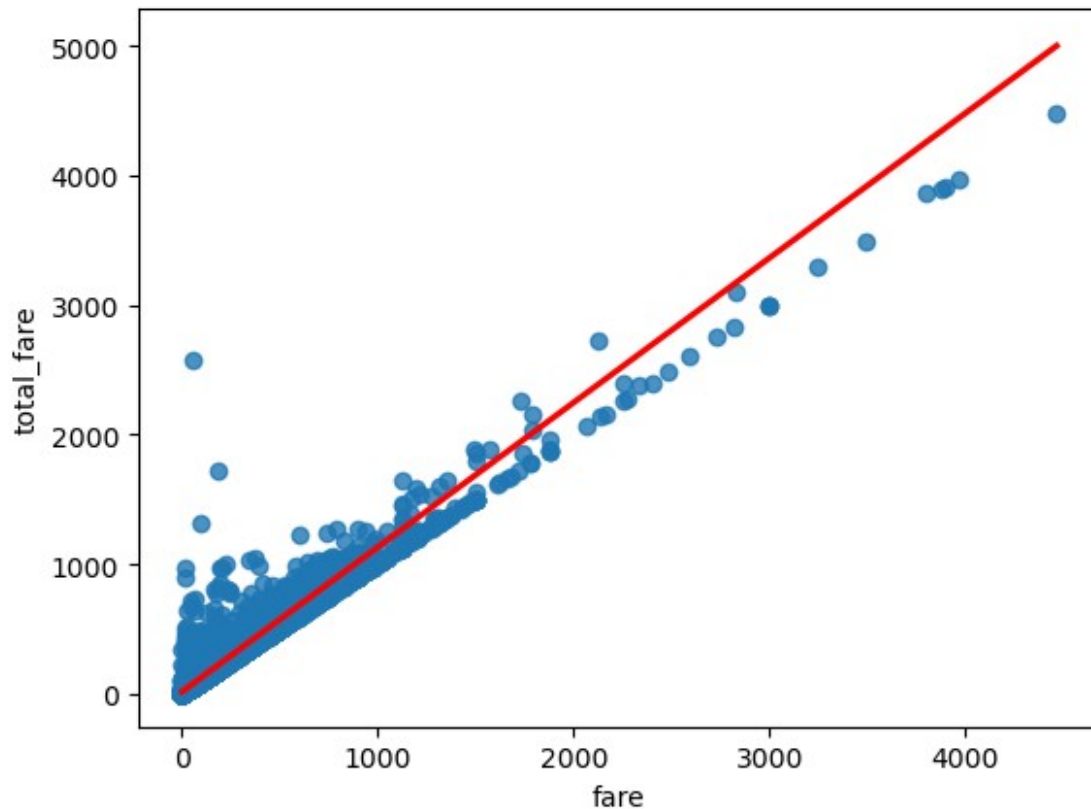


```
z.columns
```

```
Index(['trip_duration', 'distance_traveled', 'num_of_passengers',  
      'fare',  
      'tip', 'miscellaneous_fees', 'total_fare', 'surge_applied'],  
      dtype='object')
```

```
sns.regplot(x = z["fare"], y = z["total_fare"], data = z, line_kws =  
            {"color" : "red"})
```

```
<Axes: xlabel='fare', ylabel='total_fare'>
```



```
x = z[["fare", "total_fare"]]
X = x
Y = x["total_fare"]

x_train, x_test, y_train, y_test = train_test_split(X, Y, train_size =
0.7, test_size = 0.3, random_state = 100)

x_train = x_train.drop(["total_fare"], axis = 1)
x_test = x_test.drop(["total_fare"], axis = 1)

y_train = np.array(y_train).reshape(-1, 1)
y_test = np.array(y_test).reshape(-1, 1)

n = LinearRegression()
n.fit(x_train, y_train)

LinearRegression()

y_predict_train = n.predict(x_train)
r2_train = r2_score(y_true = y_train, y_pred = y_predict_train)

round(r2_train, 2)*100

93.0
```


Testing model using testing dataset

```
y_predict_test = n.predict(x_test)
r2_test = r2_score(y_true = y_test, y_pred = y_predict_test)

round(r2_test, 2)*100

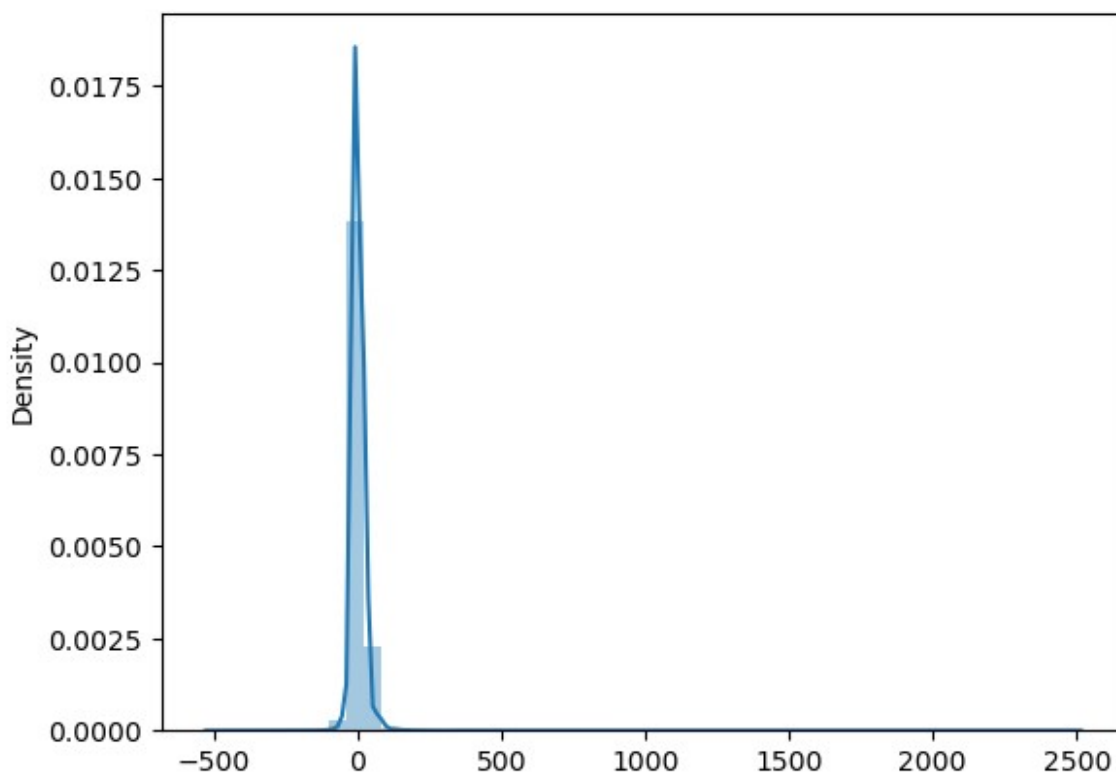
93.0

res_train = y_train - y_predict_train
res_train

array([[ -15.55867891],
       [ -12.45181161],
       [ -14.64898076],
       ...,
       [  -4.79163123],
       [ -16.36319758],
       [  -0.45229385]])

sns.distplot(res_train, kde = True)

<Axes: ylabel='Density'>
```



```
res_test = y_test - y_predict_test
```

```
res_test
array([[ -3.079825  ],
       [ -9.46520797],
       [ 16.18686863],
       ...,
       [-20.82990588],
       [ -2.41936115],
       [ -3.60897819]])

sns.distplot(res_test, kde = True)
<Axes: ylabel='Density'>
```

